

LAB 4

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Lab Session Week 4 M26

Week4 La...

Mobile Apps Development

COMP-304

Lab Session Week 4

In this lab session exercise, you will:

1. Discuss the fundamental principles and features of Material Design 3.
2. Identify and utilize various Material Design components to enhance user interfaces.
3. Apply Material Design 3 color schemes to create visually appealing and consistent app themes.
4. Design responsive UIs that adapt to large screens and foldable devices using WindowSizeClass.
5. Implement accessibility features to ensure apps are usable by all users, including those with disabilities.

Exercise 1: Introduction to Material Design 3

In this exercise, you will learn about the basics of Material Design 3 and its key features. Please follow the steps below:

1. **Understand Material Design 3**
 - o Review the principles and features of Material Design 3.
 - o Understand the importance of dynamic color systems, expanded component library, simplified typography, and improved color schemes.
2. **Explore Dynamic Color System**
 - o Learn how the dynamic color system adjusts app colors to match the user's wallpaper.
 - o Understand its support on Android 12 and above devices.

Exercise 2: Using Material Design Components

In this task, you will use various Material Design components to enhance user interfaces. Please follow the steps below:

1. **Top App Bars**
 - o Implement different types of top app bars (small, medium, large, and center-aligned).
 - o Example:

```
@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun CenterAlignedTopBar() {
    CenterAlignedTopAppBar(
        title = { Text(text = "Packt Publishing") }
    )
}
```

2. **FloatingActionButton (FAB)**
 - o Create a FloatingActionButton and ExtendedFloatingActionButton.
 - o Example:

```
FloatingActionButton(
    onClick = { /*TODO*/ },
    content = {
        Icon(
            imageVector = Icons.Default.Add,
            contentDescription = "New Chat"
        )
    }
)

ExtendedFloatingActionButton(
    onClick = { /*TODO*/ },
    content = {
        Icon(
            imageVector = Icons.Default.Add,
            contentDescription = "New Chat"
        )
        Text(
            modifier = Modifier.padding(10.dp),
            text = "New Chat"
        )
    }
)
```

3. **Bottom App Bars**
 - o Implement a BottomAppBar with icons.
 - o Example:

```
BottomAppBar(
    actions = {
        Icon(imageVector = Icons.Rounded.Home, contentDescription = "Home Screen")
        Icon(imageVector = Icons.Rounded.ShoppingCart, contentDescription = "Cart Screen")
        Icon(imageVector = Icons.Rounded.AccountCircle, contentDescription = "Account Screen")
    }
)
```

Exercise 3: Scaffold Structure in Material Design 3

In this task, you will use the Scaffold structure to support top bar, bottom bar, and FAB. Please follow the steps below:

1. **Implement Scaffold Structure**
 - o Create a Scaffold layout that includes various components.
 - o Example:

```
Scaffold(
    topBar = { CenterAlignedTopBar() },
    bottomBar = { BottomAppBar() },
    floatingActionButton = { FloatingActionButton(onClick = { /*TODO*/ }) },
    content = { paddingValues ->
        Column(
            modifier = Modifier
                .fillMaxSize()
                .padding(paddingValues)
                .background(Color.Gray.copy(alpha = 0.1f)),
            verticalArrangement = Arrangement.Center,
            horizontalAlignment = Alignment.CenterHorizontally
        ) {
            Text(
                modifier = Modifier.padding(10.dp),
                text = "Mastering Kotlin for Android Development - Chapter 4",
                textAlign = TextAlign.Center
            )
        }
    }
)
```

Exercise 4: Customizing Themes with Material Design 3

```

        content = { paddingValues ->
            Column(
                modifier = Modifier
                    .fillMaxSize()
                    .padding(paddingValues)
                    .background(Color.Gray.copy(alpha = 0.1f)),
                verticalArrangement = Arrangement.Center,
                horizontalAlignment = Alignment.CenterHorizontally
            ) {
                Text(
                    modifier = Modifier.padding(10.dp),
                    text = "Mastering Kotlin for Android Development - Chapter 4",
                    textAlign = TextAlign.Center
                )
            }
        }
    }
}

```

Exercise 4: Customizing Themes with Material Design 3

In this task, you will customize the theme of your app using Material Design 3 color schemes. Please follow the steps below:

1. Set Up Material Design 3

- o Include the Material 3 library in your project dependencies.
- o Example:

```
implementation "com.google.android.material:material:1.4.0"
```

2. Apply Color Schemes

- o Use the Material Theme Builder tool to generate theme colors.
- o Example:

```
val md_theme_light_primary = Color(0xFF006C49)
```

Exercise 5: Designing for Large Screens and Foldables

In this task, you will design responsive UIs that adapt to large screens and foldable devices. Please follow the steps below:

1. Use WindowSizeClass

- o Implement layouts suitable for varying screen sizes and orientations using WindowSizeClass.
- o Example:

```
// Example code to demonstrate WindowSizeClass usage
```

Exercise 6: Making Your App Accessible

In this task, you will implement accessibility features to ensure your app is usable by all users. Please follow the steps below:

1. Enhance Accessibility

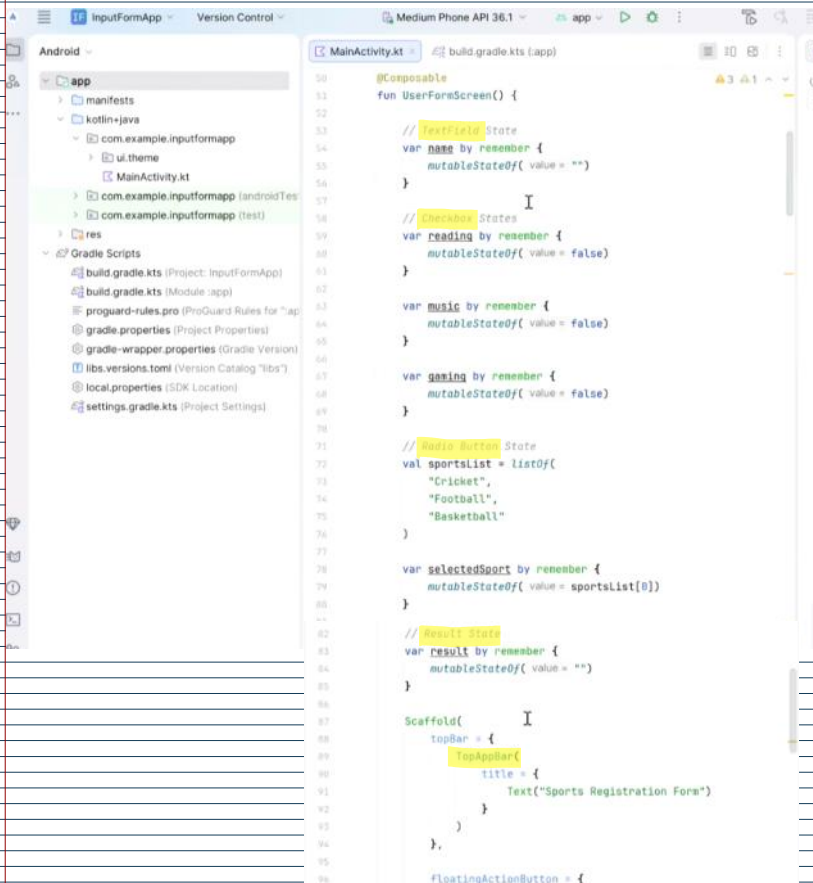
- o Use semantics to enhance accessibility.
- o Add content descriptions and ensure proper sizing for UI elements.

Demonstration

Please demonstrate the exercises at the end of the lab session. Ensure that your application runs successfully on an Android device or emulator and that all functionalities are working as expected.

Note: Use the references provided in the Week 4 lecture slides to help you with Material Design 3 and responsive UI design.

INPUT FORM APP EXAMPLE:



The screenshot shows a mobile app interface with the following sections and handwritten annotations:

- Enter Your Details**: A section header at the top.
- Name**: A text input field with the label "Name" above it. Handwritten annotation: "textfield".
- Select Hobbies**: A section header.
- Hobbies**: Three checkboxes labeled "Reading", "Music", and "Gaming". Handwritten annotation: "checkbox".
- Favorite Sport**: A section header.
- Sports**: Three radio buttons labeled "Cricket", "Football", and "Basketball". Handwritten annotation: "radio button".
- Submitted Information**: A section header.
- Click the FAB to display entered details.**: A message in a light pink box.
- FAB**: A floating action button (a circle with a plus sign) at the bottom right. Handwritten annotation: "FAB".

Handwritten notes on the right side of the image:

- "all info will be displayed here" with an arrow pointing to the "Submitted Information" section.
- "→ FAB" with an arrow pointing to the FAB button.

MainActivity.kt

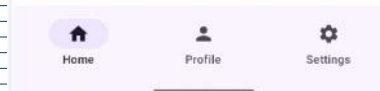
```
1 package com.example.appbarexample
```

```

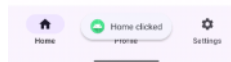
42 }
43
44
45 7 Usages
46 enum class BottomMenuItem {
47     3 Usages
48     Home,
49     2 Usages
50     Profile,
51     2 Usages
52     Settings
53 }
54
55 1 Usage
56 @OptIn(markerClass = ExperimentalMaterial3Api::class)
57 @Composable
58 fun MyApp() {
59
60     var selectedItem by remember {
61         mutableStateOf(value = BottomMenuItem.Home)
62     }
63
64     val context = LocalContext.current
65
66     Scaffold(
67         topBar = {
68             TopAppBar(
69                 title = {
70                     Text("Jetpack Compose Demo")
71                 }
72             )
73         },
74         bottomBar = {
75             BottomAppBar {
76
77                 NavigationBarItem(
78                     selected = selectedItem == BottomMenuItem.Home,
79                     onClick = {
80                         selectedItem = BottomMenuItem.Home
81
82                         Toast.makeText(
83                             context,
84                             text = "Home clicked",
85                             duration = Toast.LENGTH_SHORT
86                         ).show()
87                     },
88                     icon = {
89                         Icon(
90                             imageVector = Icons.Default.Home,
91                             contentDescription = "Home"
92                         )
93                     },
94                     label = {
95                         Text("Home")
96                     }
97                 )
98
99                 NavigationBarItem(
100                     selected = selectedItem == BottomMenuItem.Profile,
101                     onClick = {
102                         selectedItem = BottomMenuItem.Profile
103
104                         Toast.makeText(
105                             context,
106                             text = "Profile clicked",
107                             duration = Toast.LENGTH_SHORT
108                         ).show()
109                     },
110                     icon = {
111                         Icon(
112                             imageVector = Icons.Default.Person,
113                             contentDescription = "Profile"
114                         )
115                     },
116                     label = {
117                         Text("Profile")
118                     }
119                 )
120
121                 NavigationBarItem(
122                     selected = selectedItem == BottomMenuItem.Settings,
123                     onClick = {
124                         selectedItem = BottomMenuItem.Settings
125
126                         Toast.makeText(
127                             context,
128                             text = "Settings clicked",
129                             duration = Toast.LENGTH_SHORT
130                         ).show()
131                     },
132                     icon = {
133                         Icon(
134                             imageVector = Icons.Default.Settings,
135                             contentDescription = "Settings"
136                         )
137                     },
138                     label = {
139                         Text("Settings")
140                     }
141                 )
142             }
143         }
144     ) { paddingValues ->
145
146         Box(
147             modifier = Modifier

```

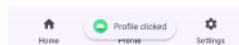
Selected: Home



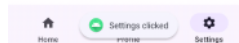
Selected: Home



Selected: Profile



Selected: Settings



```
139     }
140   ) { paddingValues ->
141
142     Box(
143       modifier = Modifier
144         .fillMaxSize()
145         .padding(paddingValues),
146       contentAlignment = Alignment.Center
147     ) {
148       Text(
149         text = "Selected: ${selectedItem.name}",
150         style = MaterialTheme.typography.headlineMedium
151       )
152     }
153   }
154 }
```